

Compiladores, 2025/2026

Trabalho prático, parte 3

Exemplos de inputs/outputs para o mooshak

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Exemplo A

Input

```
1  integer zzz;
2
3  function mainNNN() {
4      integer n;
5      bool n;
6      string s := hello("Maria");
7      string s2 := hello(5);
8      n := fun(1,2,3);
9      fun(1,2);
10     n := hello("Maria");
11     fun := 8;
12     n := misterio(n);
13     return true;
14 }
15
16 function hello( string s ) -> string {
17     return "Hello " || s;
18 }
19
20 function zzz( integer x ) -> real {
21     return x + 1;
22 }
23
24 function hello() {
25     print "Hello";
26 }
27
28 function fun( integer x, integer y ) -> integer {
29     bool b := hello;
30     return x + y;
31 }
32
33 function xpto() -> integer {
34     real r := 3.14;
35     return r;
36 }
```

Output

```
error in line 5: n already declared
error in line 7: expecting a string for argument of function hello
error in line 8: function fun expects 2 arguments
error in line 9: value of function fun must be assigned to a variable
error in line 10: operator := is invalid between integer and string
error in line 11: fun is not a variable
error in line 12: misterio not declared
error in line 13: function mainNNN does not return a value
error in line 20: zzz already declared
error in line 24: hello already declared
error in line 29: hello is not a variable
error in line 35: function xpto must return a value of type integer
error in line 37: missing main()
```

Exemplo B

Input

```
1  function main() {  
2  }  
3  
4  function hello( string s ) -> string {  
5      print "Hello " || s;  
6  }  
7  
8  function xpto() -> real {  
9      real r := 3.14;  
10     return r;  
11 }  
12  
13 function xpto2() -> integer {  
14     integer i := 5;  
15     if (i < 3)  
16         return i;  
17     print "ola";  
18 }  
19  
20
```

Output

```
error in line 6: missing return in function hello  
error in line 18: missing return in function xpto2
```

Exemplo C

Input

```
1  function sqr( real x ) -> real {
2      return x * x;
3  }
4
5  function double( integer i ) -> real {
6      return i + i;
7  }
8
9  function main() {
10     integer i := 8;
11     print sqr(i);
12     print double(i);
13 }
```

Output

```
*** Constant pool ***
*** Instructions ***
0: call 11
1: halt
2: lload -1
3: lload -1
4: dmult
5: retval 1
6: lload -1
7: lload -1
8: iadd
9: itod
10: retval 1
11: lalloc 1
12: iconst 8
13: lstore 2
14: lload 2
15: itod
16: call 2
17: dprint
18: lload 2
19: call 6
20: dprint
21: pop 1
```

```
22: ret 0
*** VM output ***
64.0
16.0
```

Exemplo C

Input

```
1  function sqr( real x ) -> real {
2      return x * x;
3  }
4
5  function double( integer i ) -> real {
6      return i + i;
7  }
8
9  function main() {
10     integer i := 8;
11     print sqr(i);
12     print double(i);
13 }
```

Output

```
*** Constant pool ***
*** Instructions ***
0: call 11
1: halt
2: lload -1
3: lload -1
4: dmult
5: retval 1
6: lload -1
7: lload -1
8: iadd
9: itod
10: retval 1
11: lalloc 1
12: iconst 8
13: lstore 2
14: lload 2
15: itod
16: call 2
17: dprint
18: lload 2
19: call 6
20: dprint
21: pop 1
```

```
22: ret 0
*** VM output ***
64.0
16.0
```

Exemplo D

Input

```
1  function sqr( integer x ) -> integer
2  {
3      return x * x;
4  }
5
6  function sqrsum( integer a, integer b ) -> integer
7  {
8      integer s;
9      s := sqr(a + b);
10     return s;
11 }
12
13 function main()
14 {
15     print sqrsum(3,2);
16 }
```

Output

```
*** Constant pool ***
*** Instructions ***
0: call 14
1: halt
2: lload -1
3: lload -1
4: imult
5: retval 1
6: lalloc 1
7: lload -2
8: lload -1
9: iadd
10: call 2
11: lstore 2
12: lload 2
13: retval 2
14: iconst 3
15: iconst 2
16: call 6
17: iprint
18: ret 0
```

```
*** VM output ***  
25
```

Exemplo E

Input

```
1  function main()
2  {
3      print fatorial(3);
4  }
5
6  function fatorial( integer n ) -> integer
7  {
8      if (n = 0) return 1;
9      return n * fatorial(n-1);
10 }
```

Output

```
*** Constant pool ***
*** Instructions ***
0: call 2
1: halt
2: iconst 3
3: call 6
4: iprint
5: ret 0
6: lload -1
7: iconst 0
8: ieq
9: jumpf 12
10: iconst 1
11: retval 1
12: lload -1
13: lload -1
14: iconst 1
15: isub
16: call 6
17: imult
18: retval 1
*** VM output ***
6
```

Exemplo F

Input

```
1  integer a := 10;
2
3  function main() {
4      integer b := 20, c := 30;
5      print a;
6      {
7          integer a := 11, c := 31;
8          {
9              integer a := 12, d := 40;
10             print a;
11             {
12                 integer e := 50;
13                 a := 13;
14                 print a;
15             }
16             print a;
17         }
18         print a;
19     }
20     print a;
21     b := 21; a := 14;
22     {
23         integer f := 60;
24         print f;
25         print a;
26     }
27 }
```

Output

```
*** Constant pool ***
*** Instructions ***
0: galloc 1
1: iconst 10
2: gstore 0
3: call 5
4: halt
5: lalloc 2
6: iconst 20
7: lstore 2
```

```
8: iconst 30
9: lstore 3
10: gload 0
11: iprint
12: lalloc 2
13: iconst 11
14: lstore 4
15: iconst 31
16: lstore 5
17: lalloc 2
18: iconst 12
19: lstore 6
20: iconst 40
21: lstore 7
22: lload 6
23: iprint
24: lalloc 1
25: iconst 50
26: lstore 8
27: iconst 13
28: lstore 6
29: lload 6
30: iprint
31: pop 1
32: lload 6
33: iprint
34: pop 2
35: lload 4
36: iprint
37: pop 2
38: gload 0
39: iprint
40: iconst 21
41: lstore 2
42: iconst 14
43: gstore 0
44: lalloc 1
45: iconst 60
46: lstore 4
47: lload 4
48: iprint
49: gload 0
50: iprint
51: pop 1
52: pop 2
53: ret 0
*** VM output ***
10
```

12
13
13
11
10
60
14

Exemplo G

Input

```
1  function xpto() -> integer {
2      integer x, y;
3      {
4          integer a, b, c;
5          print "0i";
6          return 1;
7      }
8      print "Ai";
9  }
10
11 function main() {
12     integer x;
13     x := xpto();
14 }
```

Output

```
*** Constant pool ***
0: "0i"
1: "Ai"
*** Instructions ***
0: call 12
1: halt
2: lalloc 2
3: lalloc 3
4: sconst 0
5: sprint
6: iconst 1
7: retval 0
8: pop 3
9: sconst 1
10: sprint
11: pop 2
12: lalloc 1
13: call 2
14: lstore 2
15: pop 1
16: ret 0
*** VM output ***
```

Oi